Linear Algebra
(113-2, 2025 Spring)

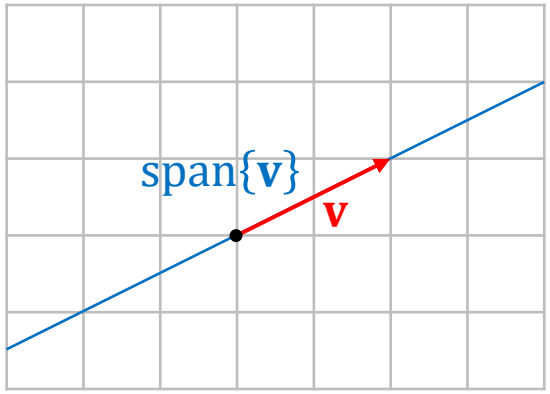
LINEAR INDEPENDENCE 1.7

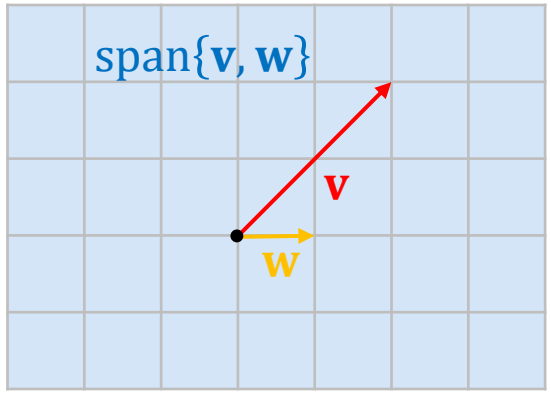
SUBSPACES 2.8

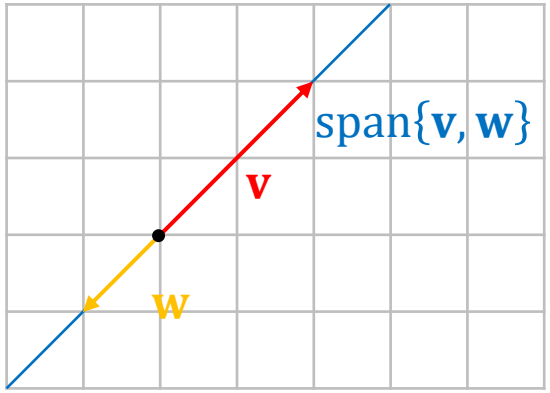
Linear Independence & Subspaces

- Linear Independence (Sec. 1.7)
- Subspaces (Sec. 2.8)

Recall: span of vectors in $\mathbb{R}^2$







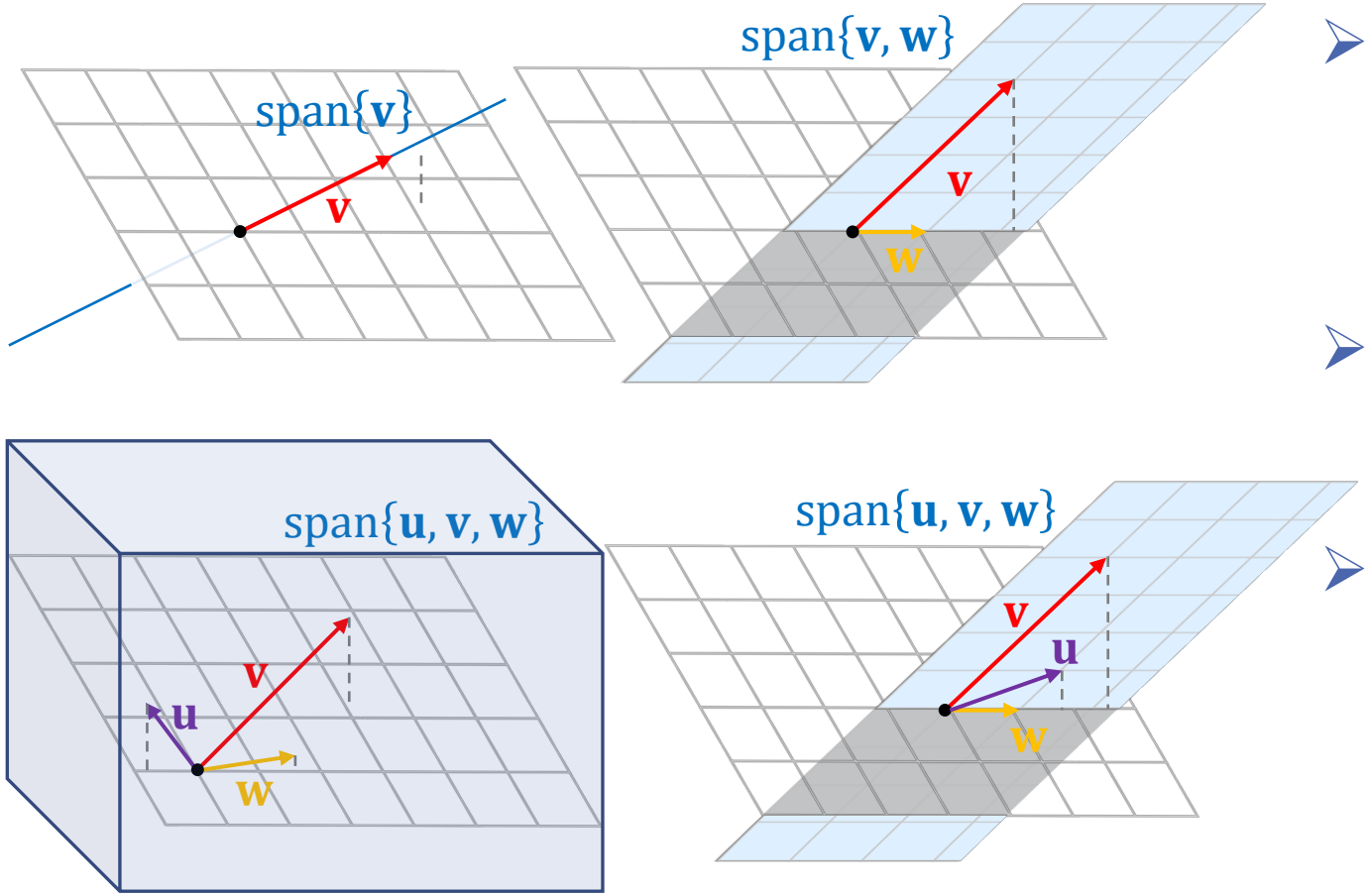
Sometimes, the span set consists of fewer vectors than you expect.

w: redundant vector

- e.g. the left-bottom figure has a span set $span\{\mathbf{v}\}$ or $span\{\mathbf{w}\}$ instead of $span\{\mathbf{v}, \mathbf{w}\}$.
- This means that (at least) one of the vectors is redundant: "too many" vectors to describe the span.
- Note: If one vector is in the span set of other vectors, the vector does not make the span bigger.

$$\mathbf{w} \in span\{\mathbf{v}\}$$
$$\Leftrightarrow \mathbf{w} \text{ is a scalar multiple of } \mathbf{v}.$$

Recall: span of vectors in $\mathbb{R}^3$



- e.g. the right-bottom figure has a span set $span\{\mathbf{u}, \mathbf{v}\}$ or $span\{\mathbf{u}, \mathbf{w}\}$ or $span\{\mathbf{v}, \mathbf{w}\}$ instead of $span\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$.
- This means that (at least) one of the vectors is redundant: "too many" vectors to describe the span.
- Note: If one vector is in the span set of other vectors, the vector does not make the span bigger.

$$\mathbf{u} \in span\{\mathbf{v}, \mathbf{w}\}$$

$\Leftrightarrow$ $\mathbf{u}$ is a linear combination of $\mathbf{v}, \mathbf{w}$.

Linear Independence

Definition:

LI

An indexed set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_p\mathbf{v}_p = \mathbf{0}$$

LD

has only the trivial solution. The set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is said to be **linearly dependent** otherwise.

- $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is linearly dependent if there exist weights $\boldsymbol{c_1}, \dots, \boldsymbol{c_p}$ **not all zero**, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p = \mathbf{0}.$$

  This is called a **linear dependence relation**.

Check Linear Independence

Example 1:

Let $\mathbf{v}_1 = \begin{bmatrix}1\\2\\3\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}4\\5\\6\end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix}2\\1\\0\end{bmatrix}$.

a. Determine if the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.

b. If possible, find a linear dependence relation among $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

Determine if there exists a **nontrivial** solution of equation: $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + x_3\mathbf{v}_3 = \mathbf{0}$

let $\vec{v}_1 + \vec{v}_2 + \vec{v}_3 = \vec{0}$

x_3 is free ∞ sols

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{array}\right] \begin{matrix} \\ -2 \\ -3 \end{matrix} \sim \left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & -6 & -6 & 0 \end{array}\right] \begin{matrix} \\ \div -3 \\ -2 \end{matrix} \sim \left[\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{matrix} -4 \\ \\ \\ \end{matrix} \sim \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Not only trivial solution : $\alpha \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, $\alpha \in \mathbb{R}$

let $x_3 = \alpha$

a. linearly dependent, not linearly independent

b. let $x_3 = \alpha = 1$

$\vec{x} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \Rightarrow 2\vec{v}_1 - \vec{v}_2 + \vec{v}_3 = \vec{0}$

Check Linear Independence (Cont'd)

Example 1:

Let $\mathbf{v}_1 = \begin{bmatrix}1\\2\\3\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}4\\5\\6\end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix}2\\1\\0\end{bmatrix}$.

a. Determine if the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.

b. If possible, find a linear dependence relation among $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

Determine if there exists a **nontrivial** solution of equation: $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + x_3\mathbf{v}_3 = \mathbf{0}$

$$\left[\begin{array}{ccc|c}1 & 4 & 2 & 0\\2 & 5 & 1 & 0\\3 & 6 & 0 & 0\end{array}\right] \sim \left[\begin{array}{ccc|c}1 & 4 & 2 & 0\\0 & -3 & -3 & 0\\0 & 0 & 0 & 0\end{array}\right]$$

$\Rightarrow x_3$ is a free variable
and $x_2 = -x_3, x_1 = 2x_3$

$\Rightarrow$ The system has many solutions

$\Rightarrow$ nontrivial solutions exists

$\Rightarrow \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent.

linear dependence relation:

$$2x_3\mathbf{v}_1 - x_3\mathbf{v}_2 + x_3\mathbf{v}_3 = \mathbf{0}$$

(take $x_3 = 1$) $\Rightarrow 2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$

Linear Independence & Matrix Columns

In general, $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is linearly independent

$\Leftrightarrow x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_p\mathbf{v}_p = \mathbf{0}$ has only the trivial solution. No free variable

$\Leftrightarrow A\mathbf{x} = \mathbf{0}$ has only the trivial solution where $\mathrm{A} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_p]$.

$\Leftrightarrow A$ has a pivot in each column.

Columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ of A are linearly independent.

- Solving the matrix equation $A\mathbf{x} = \mathbf{0}$ will either verify that the columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ of A are **linearly independent**, or will produce a **linear dependence relation**.

- A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is linearly dependent if and only if **one of the vectors is in the span of the other ones.**

e.g. $2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$

$\Rightarrow \mathbf{v}_3 = -2\mathbf{v}_1 + \mathbf{v}_2$
$\Rightarrow \mathbf{v}_3$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2$.

$\vec{v}_3 \in \text{span}\{\vec{v}_1, \vec{v}_2\} = \{\alpha\vec{v}_1, \beta\vec{v}_2\}$

Linear Dependence Characterization

THEOREM 7

Characterization of Linearly Dependent Sets

An indexed set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then some $\mathbf{v}_j$ (with $j > 1$) is a linear combination of the preceding vectors, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{j-1}$.

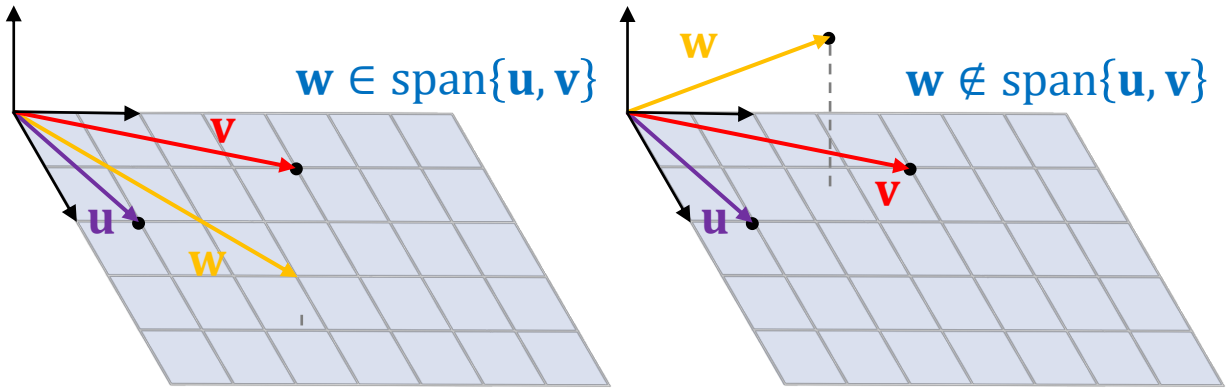
- If $S = \{\mathbf{v}_1\}$, then $\mathbf{0} = x_1\mathbf{v}_1$ has only trivial solution when $\mathbf{v}_1 \neq \mathbf{0}$.
  $\Rightarrow S$ is linearly independent.
- If $S = \{\mathbf{0}\}$, then S is linearly dependent since $\mathbf{0} = x_1\mathbf{0}$ has nontrivial solution.
- If $S = \{\mathbf{v}_1, \mathbf{v}_2\}$: linearly dependent, then $\mathbf{v}_2 = \alpha\mathbf{v}_1$.
- If $S = \{\mathbf{v}_1, \mathbf{v}_2\}$: linearly independent, then $\mathbf{v}_2 \neq \alpha\mathbf{v}_1$.
- If $\mathbf{v}_1 = c_2\mathbf{v}_2 + c_3\mathbf{v}_3$, then $\mathbf{0} = (-1)\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + 0\mathbf{v}_4 + \cdots + 0\mathbf{v}_p$
  $\Rightarrow S$: linearly dependent.

Linear Dependence Characterization (Cont'd)

Example 4:

Let $\mathbf{u} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix}$.

Describe the set spanned by $\mathbf{u}$ and $\mathbf{v}$, and explain why a vector $\mathbf{w}$ is in $span\{\mathbf{u}, \mathbf{v}\}$ if and only if $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly dependent.



Sol.

$\mathbf{u} \neq t\mathbf{v} \Rightarrow \mathbf{u}$ is not a multiple of $\mathbf{v}$.

$$span\{\mathbf{u}, \mathbf{v}\} = \{\alpha\mathbf{u} + \beta\mathbf{v}\}$$
$$= \left\{ \alpha \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 6 \\ 0 \end{bmatrix} \middle| \alpha, \beta \in \mathbb{R} \right\}$$

$\Rightarrow span\{\mathbf{u}, \mathbf{v}\}$ is a plane in $\mathbb{R}^3$.

- If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ linearly dependent and $\mathbf{u} \neq t\mathbf{v}$, then $\mathbf{w}$ must be a linear combination of $\mathbf{u}, \mathbf{v}$, i.e. $\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v}$.
  $\Rightarrow \mathbf{w} \in span\{\mathbf{u}, \mathbf{v}\}$.
- If $\mathbf{w}$ is in the plane spanned by $\mathbf{u}, \mathbf{v}$, then $\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v}$ with some α and β.
  $\Rightarrow \{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly dependent.

$\alpha\vec{u} + \beta\vec{v} - \vec{w} = 0$ has not only trivial solution

Linear Dependence Characterization (Cont'd)

Warning:
Theorem 7 does *not* say that every vector in a linearly dependent set is a linear combination of the preceding vectors.
A vector in a linearly dependent set may fail to be a linear combination of the other vectors.

eg. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right\}$: LD

$\vec{v}_3 = 2\vec{v}_2$ $\vec{v}_3$ is a multiple of $\vec{v}_2$,

but $\vec{v}_1$ is not a linear combination of $\vec{v}_2$ and $\vec{v}_3$.

Practice Problems 1:

Let $\mathbf{u} = \begin{bmatrix} 3 \\ 2 \\ -4 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -6 \\ 1 \\ 7 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ -5 \\ 2 \end{bmatrix}$, and $\mathbf{z} = \begin{bmatrix} 3 \\ 7 \\ -5 \end{bmatrix}$.

a. Are the sets $\{\mathbf{u}, \mathbf{v}\}$ LI, $\{\mathbf{u}, \mathbf{w}\}$ LI, $\{\mathbf{u}, \mathbf{z}\}$ LI, $\{\mathbf{v}, \mathbf{w}\}$ LI, $\{\mathbf{v}, \mathbf{z}\}$ LI, and $\{\mathbf{w}, \mathbf{z}\}$ LI each linearly independent? Why or why not? scalar multiple

All sets are linearly independent, since none of them are multiple of others

b. Does the answer to Part (a) imply that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{z}\}$ is linearly independent? No, you only know the relations of 1 on 1 replacements

c. To determine if $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{z}\}$ is linearly dependent, is it wise to check if, say, $\mathbf{w}$ is a linear combination of $\mathbf{u}, \mathbf{v}$, and $\mathbf{z}$? a is not a wise way but a quick check

d. Is $\{\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{z}\}$ linearly dependent?

where $A = [\vec{u}, \vec{v}, \vec{w}, \vec{z}] \Rightarrow A\vec{x} = 0$ has the only trivial solution

Linear Dependence

THEOREM 8

If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

$$\begin{array}{cc} & p \\ n & \begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix} \end{array} \sim \begin{bmatrix} * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & * & * & * \end{bmatrix}$$

⇒ # variables (p) > #equations (n)

⇒ The linear system must have free variables.

⇒ The linear system has a nontrivial solution.

⇒ The columns of A are linearly dependent.

- e.g. $[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

  $n = 2, p = 3 \Rightarrow p > n \Rightarrow$linearly dependent.

- e.g. $[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] = \begin{bmatrix} 2 & 4 & -2 \\ 1 & -1 & 2 \end{bmatrix}$

  $n = 2, p = 3 \Rightarrow p > n \Rightarrow$linearly dependent.

Linear Dependence

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If a set contains more vectors than there are entries in each vector, then the set is linearly dependent. That is, any set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

$$\begin{array}{cc} & p \\ n & \begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix} \end{array} \sim \begin{bmatrix} * & * & * & * & * \\ 0 & * & * & * & * \\ 0 & 0 & * & * & * \end{bmatrix}$$

⇒ # variables (p) > #equations (n)

⇒ The linear system must have free variables.

⇒ The linear system has a nontrivial solution.

⇒ The columns of A are linearly dependent.

- e.g. $[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \mathbf{v}_4] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

  $n = 3, p = 4 \Rightarrow p > n \Rightarrow$linearly dependent.

- e.g. $[\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \mathbf{v}_4] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

  $n = 3, p = 4 \Rightarrow p > n \Rightarrow$linearly dependent.

Linear Dependence (Cont'd)

THEOREM 9

If a set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ contains the zero vector, then the set is linearly dependent.

If $\mathbf{v}_1 = \mathbf{0}$, there is a linear combination $1\mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_p = \mathbf{0}$.
$\Rightarrow S$ is linearly dependent.

Exercise 11 (P.89)

$\left\{\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix}, \begin{bmatrix} -1 \\ 5 \\ h \end{bmatrix}\right\}$: linearly dependent. Find h.

① $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 = \mathbf{0}$

② $\mathbf{v}_3 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$

$$\left[\begin{array}{ccc|c} 1 & 3 & -1 & 0 \\ -1 & -5 & 5 & 0 \\ 4 & 7 & h & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 3 & -1 & 0 \\ 0 & -2 & 4 & 0 \\ 0 & -5 & h+4 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 3 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & h-6 & 0 \end{array}\right]$$

Linearly Independent/Dependent

Example 6:

Check linear dependence:

a) $\begin{bmatrix}1\\7\\6\end{bmatrix}, \begin{bmatrix}2\\0\\9\end{bmatrix}, \begin{bmatrix}3\\1\\5\end{bmatrix}, \begin{bmatrix}4\\1\\8\end{bmatrix}$

b) $\begin{bmatrix}2\\3\\5\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\8\end{bmatrix}$

c) $\begin{bmatrix}-2\\4\\6\\10\end{bmatrix}, \begin{bmatrix}3\\-6\\-9\\15\end{bmatrix}$

Linear Independence & Subspaces

- Linear Independence (Sec. 1.7)
- Subspaces (Sec. 2.8)

Subspaces of $\mathbb{R}^n$

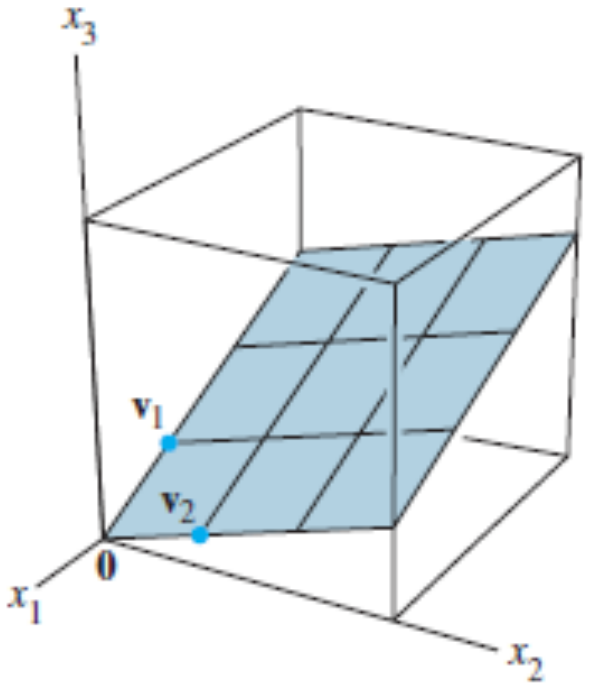
Definition:

A subspace of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:

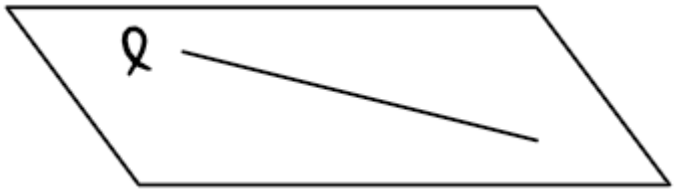
a) The zero vector is in H.

b) For each $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u} + \mathbf{v}$ is in H.

c) For each $\mathbf{u}$ in H and each scalar c, the vector $c\mathbf{u}$ is in H.



Q. A line $l \subset \mathbb{R}^2$. Is l a subspace of $\mathbb{R}^2$?



- A plane through the origin is the standard way to visualize the subspace.

Subspaces of $\mathbb{R}^n$ (Cont'd)

Example 1:

If $\mathbf{v}_1$ and $\mathbf{v}_2$ are in $\mathbb{R}^n$ and $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$, then H is a subspace of $\mathbb{R}^n$.

Sol.

Since $H = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\} = \{\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 | \alpha, \beta: \text{scalar}\}$,

Check

1. $\mathbf{0} \in H$?
2. $\text{u}, \text{v} \in H \Rightarrow \alpha\mathbf{u} + \beta\mathbf{v} \in H$?

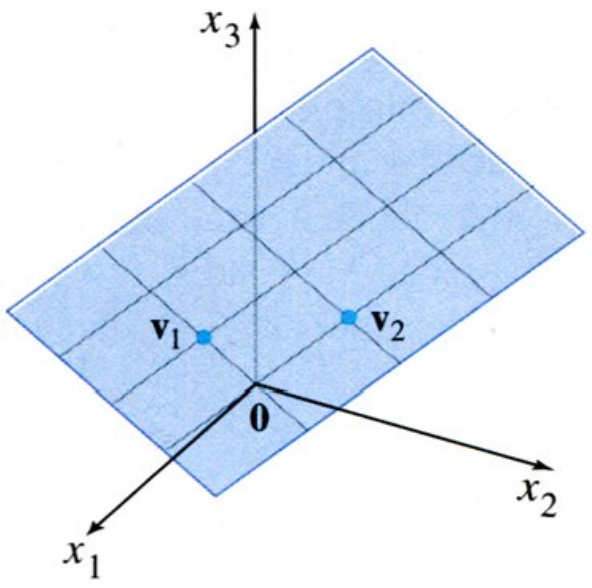


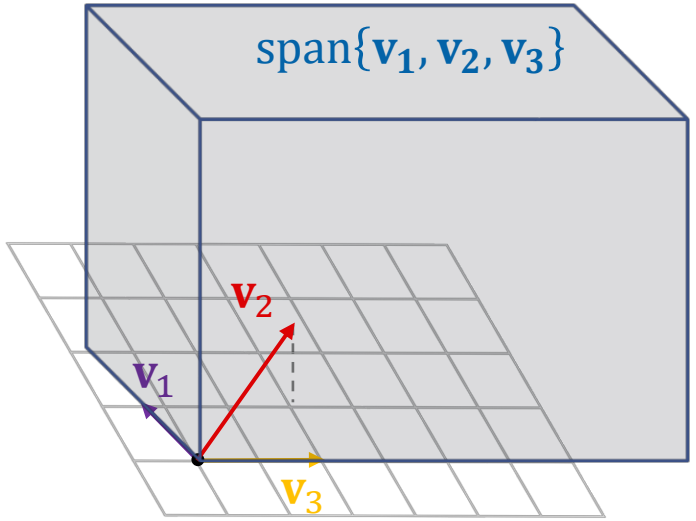
FIGURE 1

Span $\{\mathbf{v}_1, \mathbf{v}_2\}$ as a plane through the origin.

Subspaces of $\mathbb{R}^n$ (Cont'd)

For $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p \in \mathbb{R}^n$,

- $H = Span\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is the subspace spanned by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$.
  a) The zero vector is in H. ✔
  b) For each $\mathbf{v}_i, \mathbf{v}_j$ $(i \neq j)$ in H, the sum $\mathbf{v}_i + \mathbf{v}_j$ is in H. ✔
  c) For each $\mathbf{v}_i$ in H and each scalar c, the vector $c\mathbf{v}_i$ is in H. ✔



Column Space & Null Space of A Matrix

Definition:

1) The **column space** of a matrix A is the set $col(A)$ of all linear combinations of the columns of A.

2) The **null space** of a matrix A is the set $nul(A)$ of all solutions of the homogenous equation $A\mathbf{x} = \mathbf{0}$.

Given $A = [\boldsymbol{a}_1 \quad \boldsymbol{a}_2 \quad \dots \quad \boldsymbol{a}_n] : m \times n$,

- $col(A)$
  $= \{\alpha_1 \boldsymbol{a}_1 + \alpha_2 \boldsymbol{a}_2 + \cdots + \alpha_n \boldsymbol{a}_n | \forall \alpha_i \in \mathbb{R}\}$
  $= span(A)$

- $nul(A) = \{\mathbf{x} \in \mathbb{R}^n | A\mathbf{x} = \mathbf{0}\}$

Q. Given a vector $\mathbf{b}$, is $\mathbf{b}$ in $nul(A)$?

Column Space

Example 4:

Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$.

Determine whether $\mathbf{b}$ is in the column space of A.

Column Space & Null Space of A Matrix (Cont'd)

Given a matrix $A: m \times n$,

a) Column space is a subspace of $\mathbb{R}^m$.

b) Null space is a subspace of $\mathbb{R}^n$. (Theorem 12)

→ Assume that $A = [\boldsymbol{a}_1 \quad \boldsymbol{a}_2 \quad \dots \quad \boldsymbol{a}_n]$, $\boldsymbol{a}_i \in \mathbb{R}^m, \forall i = 1,2,\dots,n$

a) $col(A) = \text{span}\{\boldsymbol{a}_1, \boldsymbol{a}_2, \dots, \boldsymbol{a}_n\}$
$= \text{span}(A)$

1) …
2)

b) See Theorem 12.

1) $\mathbf{0} = 0 \cdot \boldsymbol{a}_1 + 0 \cdot \boldsymbol{a}_2 + \cdots + 0 \cdot \boldsymbol{a}_n \in col(A)$

2) Let $\mathbf{u} = \alpha_1 \boldsymbol{a}_1 + \alpha_2 \boldsymbol{a}_2 + \cdots + \alpha_n \boldsymbol{a}_n$ *and*
$\mathbf{v} = \beta_1 \boldsymbol{a}_1 + \beta_2 \boldsymbol{a}_2 + \cdots + \beta_n \boldsymbol{a}_n$

$$\begin{aligned}
&\alpha\mathbf{u} + \beta\mathbf{v} \\
&= \alpha(\alpha_1 \boldsymbol{a}_1 + \alpha_2 \boldsymbol{a}_2 + \cdots + \alpha_n \boldsymbol{a}_n) \\
&+\beta(\beta_1 \boldsymbol{a}_1 + \beta_2 \boldsymbol{a}_2 + \cdots + \beta_n \boldsymbol{a}_n) \\
&= (\alpha\alpha_1 + \beta\beta_1)\boldsymbol{a}_1 + (\alpha\alpha_2 + \beta\beta_2)\boldsymbol{a}_2 + \cdots \\
&+(\alpha\alpha_n + \beta\beta_n)\boldsymbol{a}_n \\
&\in col(A)
\end{aligned}$$

a linear combination of the columns of *A*.

From 1 and 2, $col(\mathrm{A})$ is a subspace of $\mathbb{R}^m$.

Null Space Theorm

THEOREM 12

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions of a system $A\mathbf{x} = \mathbf{0}$ of m homogenous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

pf.

1. Check if $0 \in nul(A)$.

$$nul(A) = \{\mathbf{x} \in \mathbb{R}^n | A\mathbf{x} = \mathbf{0}\}$$
$$A \cdot \mathbf{0}_{n\times 1} = \mathbf{0}_{m\times 1} \Rightarrow \mathbf{0} \in nul(A)$$

2. Check if

$\alpha\mathbf{u} + \beta\mathbf{v} \in nul(\mathbf{A})$, for $\mathbf{u}, \mathbf{v} \in nul(A) \subset \mathbb{R}^n$.

Let $\mathbf{u}, \mathbf{v} \in nul(A) \Rightarrow A\mathbf{u} = \mathbf{0}\ \boldsymbol{and}\ A\mathbf{v} = \mathbf{0}$.

Let $\alpha, \beta \in \mathbb{R} \rightarrow \alpha\mathbf{u} + \beta\mathbf{v} = \mathbf{p}$

$$A \cdot \mathbf{p} = A(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha A\mathbf{u} + \beta A\mathbf{v}$$
$$= \alpha \cdot \mathbf{0} + \beta \cdot \mathbf{0} = \mathbf{0}$$
$$\Rightarrow \mathbf{p} = \alpha\mathbf{u} + \beta\mathbf{v} \in nul(A)$$

According to 1 and 2, $nul(A)$ is a subspace of $\mathbb{R}^n$.

Basis for a Subspace

Definition:

A basis for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H.

- e.g. $\left\{ \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{n\times 1}, \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}_{n\times 1}, \dots, \mathbf{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}_{n\times 1} \right\}$ is a basis for $\mathbb{R}^n$.

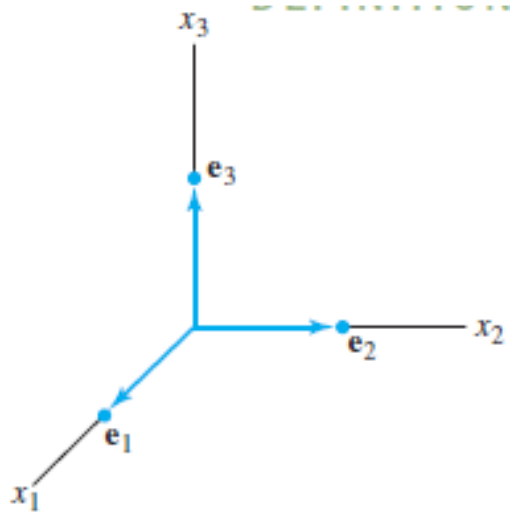


FIGURE 3
The standard basis for $\mathbb{R}^3$.

- $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$: standard basis for $\mathbb{R}^n$.

  Given $S = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3 | \mathbf{b}_i \in \mathbb{R}^3\}$: L. I. that S is in a subspace H spanned by S, i.e.,
  $H = span(S) = span\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$.
  $\Rightarrow S = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ is a basis for the subspace H of $\mathbb{R}^3$.

Basis for a Subspace (Cont'd)

THEOREM 13

The pivot columns of a matrix A form a basis for the column space of A.

Example 6:

Find a basis for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}.$$

Basis for a Subspace (Cont'd)

THEOREM 13

The pivot columns of a matrix *A* form a basis for the column space of *A*.

Example 7:

Find a basis for $col(B)$.

$$\mathrm{B} = \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Example 8:

Find a basis $col(\mathrm{A})$.

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ -2 & -2 & 2 & -8 & 2 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix} \sim B.$$

Basis for a Subspace (Cont'd)

[Warning]:

$\mathbf{A} \sim \mathbf{B}$: echelon form.

Find a basis for $col(\mathbf{A})$

1. Row reduce **A** to echelon form **B**.
2. Determine the pivot columns.
3. Collect the corresponding pivot columns in **A** to form the basis.

→ Be careful to use the pivot columns of **A**. Do NOT use **B**.

Assignment 04

1.7 – 2,6,10,38

2.8 – 2,12,26

Due 3/25 Tue. In-class